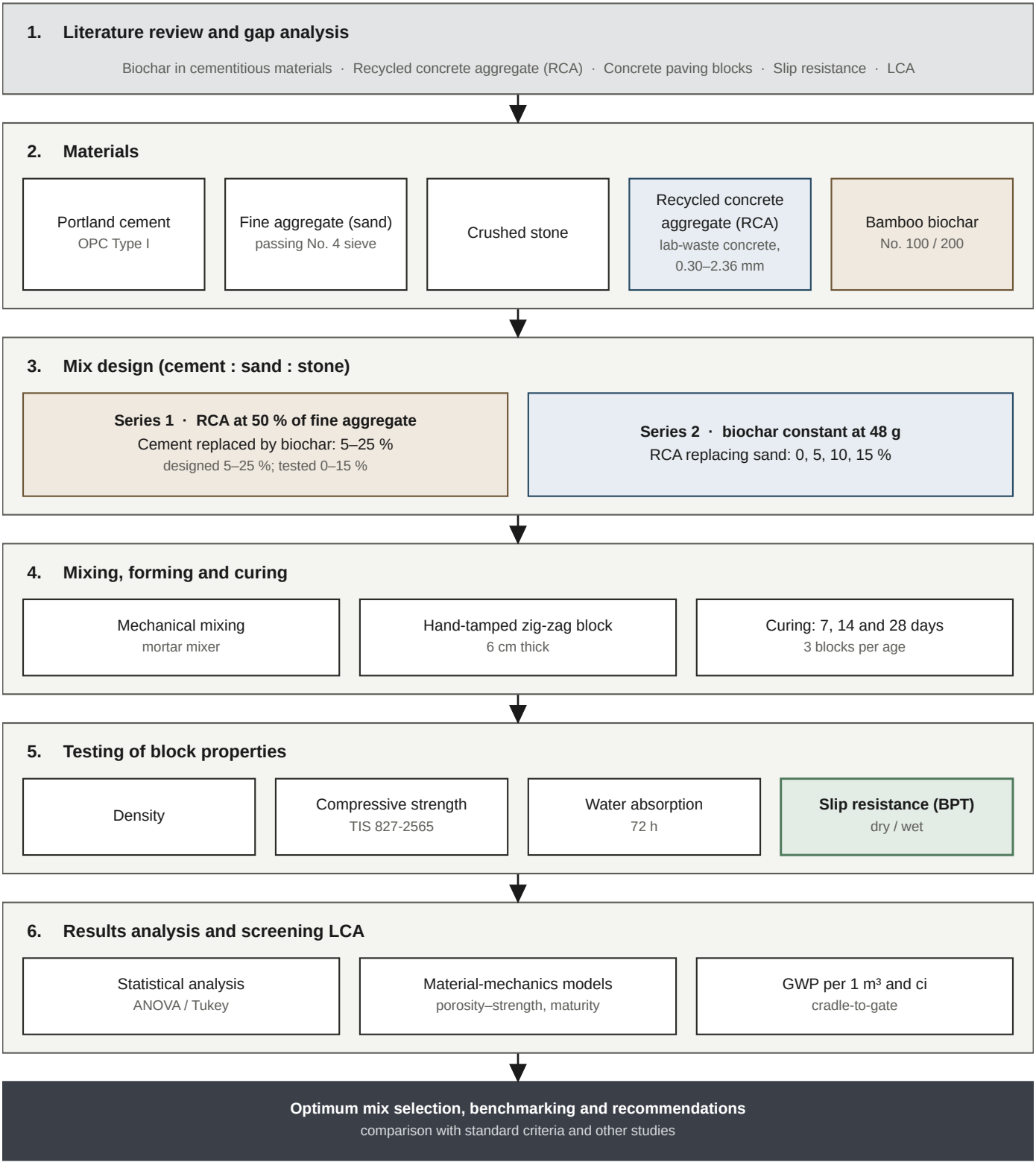
1. Literature review and gap analysis
Biochar in cementitious materials · Recycled concrete aggregate (RCA) · Concrete paving blocks · Slip resistance · LCA
2. Materials
Portland cement
OPC Type I
Fine aggregate (sand)
passing No. 4 sieve
Crushed stone
Recycled concrete aggregate (RCA)
lab-waste concrete, 0.30–2.36 mm
Bamboo biochar
No. 100 / 200
3. Mix design (cement : sand : stone)
Series 1 · RCA at 50 % of fine aggregate
Cement replaced by biochar: 5–25 %
designed 5–25 %; tested 0–15 %
Series 2 · biochar constant at 48 g
RCA replacing sand: 0, 5, 10, 15 %
4. Mixing, forming and curing
Mechanical mixing
mortar mixer
Hand-tamped zig-zag block
6 cm thick
Curing: 7, 14 and 28 days
3 blocks per age
5. Testing of block properties
Density
Compressive strength
TIS 827-2565
Water absorption
72 h
Slip resistance (BPT)
dry / wet
6. Results analysis and screening LCA
Statistical analysis
ANOVA / Tukey
Material-mechanics models
porosity–strength, maturity
GWP per 1 m³ and ci
cradle-to-gate
Optimum mix selection, benchmarking and recommendations
comparison with standard criteria and other studies